

An Intelligent Interactive Visual Analytics System for Exploring Large and Multi-Scale Pathology Images

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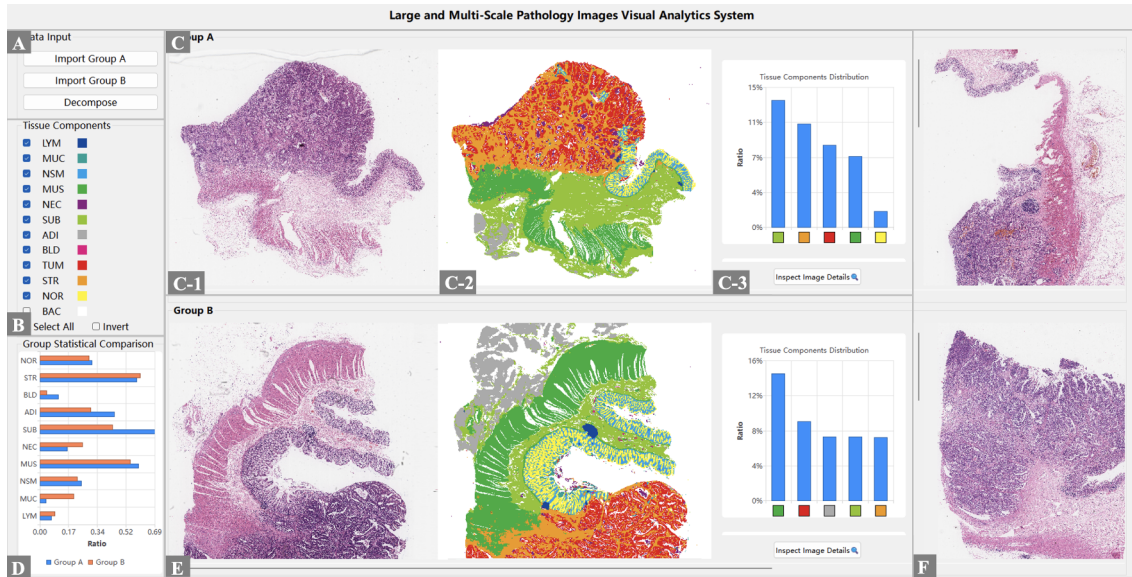


Fig. 1: The interface of our intelligent interactive visual analytics system for pathology images involves (A) data input and pathology image segmentation module, (B) tissue component selection panel, (C) subject-level comparison featuring tri-view visualization (C-1 original images, C-2 segmented annotations, C-3 component distributions), and (D-E) cohort-level analysis dashboard incorporating (D) group statistical comparison barchart and (E) synchronized multi-image sliding window.

Abstract—Pathology images are crucial for cancer diagnosis and treatment. Although artificial intelligence has driven rapid advancements in pathology image analysis, the interpretation of ultra-large and multi-scale pathology images in clinical practice still heavily relies on physicians' experience. Clinicians need to repeatedly zoom in and out on individual slides to compare and assess pathological details — a process that is both time-consuming and prone to visual fatigue. The system first employs a diffusion model to perform tissue segmentation on pathology images, then calculates pathological tissue proportions and morphological metrics. Finally, through multi-scale dynamic comparison and multi-level visual evaluation, the system facilitates comprehensive and precise analysis of pathology images. The system provides clinicians with an intelligent and interactive tool for pathology image interpretation, enabling efficient visualization and precise analysis of pathological details, thereby reducing the effort require for detailed analysis.

Index Terms—Pathology Image, Diffusion Model, Large-Scale, Visual Analytics, Interactive Exploration

1 INTRODUCTION

Pathology images serve as crucial references for cancer diagnosis, recurrence risk assessment, and treatment plan selection. With advances in artificial intelligence, numerous diagnostic algorithms and models for pathology image analysis have been developed and validated, demonstrating successful applications across multiple cancer types [7, 27]. AI-powered pathological diagnosis foundation models have achieved groundbreaking progress in critical tasks including cancer subtype classification and histological grading, offering a robust intelligent solution for complex pathological diagnosis and analytical workflows [15]. Nevertheless, pathology images contain intricate and abundant information. Relying solely on AI to analyze and localize lesion features cannot meet clinicians' demands for fine-grained interpretation and multi-level comparison of ultra-large-scale pathology images. In clinical practice, current cancer diagnosis and malignancy grading still heavily depend on physicians' expertise, particularly for cancer patients requiring surgical resection, where accurate and intuitive analysis of ultra-large-scale

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pathology images is especially important.

In clinical pathology image analysis, experts often need to perform slide-by-slide examination, frequent zooming and comparison of pathology images to enable meticulous observation at varying scales, thereby facilitating accurate disease assessment. However, this work process faces three major challenges: First, ultra-large-scale pathology images typically contain millions of pixels, demanding substantial computational resources for processing and analysis. Efficiently extracting critical pathological information from such high-resolution images remains a pressing issue. Second, pathology images exhibit multi-scale tissue characteristics, where different regions of tumors and their microenvironments may display distinct morphological and microstructural patterns. Conducting effective multi-scale image analysis poses a significant challenge. Third, pathology images usually consist of multiple layers, each encoding different types of biological and pathological information (e.g., tissue morphology). Precise layer decomposition and feature extraction are pivotal to enhancing analytical accuracy. Thus, there is an urgent clinical need for rapid and precise image deconstruction coupled with efficient comparative analysis of pathology images.

Visual analytics is a human-in-the-loop framework that synthesizes data mining algorithms and interactive visualization techniques. This approach translates raw pathology images into interpretable visual contexts, empowering pathologists to collaboratively decode complex pathological tissues. Additionally, visualization methods can integrate AI analysis results to improve the efficiency of pathological analysis and to bolster the scientific precision of clinical evaluations by experts.

Building on these considerations, this paper introduces an intelligent visual analytics system tailored for the analysis of pathology images. Initially, a diffusion model is utilized to segment the pathology images and to extract information pertaining to the pathological tissue components. Subsequently, the system computes the proportions of pathological tissues and morphological indicators. Following this, the system facilitates efficient visualization and precise analysis of pathology images through multi-level interactive exploration, thereby enhancing clinical evaluations with more informative insights.

- An intelligent visual analytics system for pathology images, specializing in tissue segmentation and multi-scale comparative analysis of large-scale pathology images.
- A diffusion model-based method that enables in-depth analysis and efficient interpretation of pathological tissue.
- Experimental validation on real-world cancer pathology datasets, demonstrating the system's practical utility.

2 RELATED WORK

2.1 Pathology Image Visualization

Pathology image visualization is a critical research direction in medical image processing, aiming to assist medical experts in effectively analyzing and interpreting histopathology images through visualization techniques. Warchol et al. introduced *Pseudo*, a system that enhances visualization of multi-channel biomedical imaging data through optimized color palettes, enabling more accurate perception of spatial relationships between channels [24]. Marini et al. developed a Python library for multi-scale pathology image analysis, leveraging CNN-based classification and segmentation of whole-slide images [16]. Molin et al. proposed *Scale Stain*, a novel visualization method that enhances multi-resolution features in pathology images by integrating color deconvolution with maximum intensity projection, thereby improving visualization of microscopic structures [17]. Horák et al. developed *xOpat*, an open-source pathology analysis tool offering flexible data layer rendering and visualization configurations [11]. Duan et al. created *SenseCare*, a medical imaging informatics platform with interactive 3D visualization capabilities, integrating AI toolkits for multi-modal image segmentation, registration, lesion detection, and landmark identification [22]. Khaled et al. presented *HistoContours*, a framework for end-to-end analysis of whole-slide histopathology images using deep learning-based nuclear localization, classification,

and color-coded isocontour visualization [2]. Jessup et al. designed *Scope2Screen*, a system employing focus+context techniques to analyze multivariate image features, cell types, and spatial arrangements across scales for pathological tumor assessment [12]. Warchol et al. also introduced *Visinity*, a visualization system for multiplexed tissue imaging data that deciphers cell interaction patterns via spatial neighborhood analysis [23]. Khvostikov and colleagues developed *PathScribe*, a software tool designed to facilitate the visualization and analysis of high-resolution whole-slide histology images across a variety of formats [13]. Falk et al. implemented an interactive visualization system using raycasting for histopathology data exploration at native resolution [8].

Despite significant advancements in pathology image visualization, clinical practice still lacks robust multi-scale comparative analysis techniques for ultra-large, ultra-high-resolution pathology images. To address this gap, we developed a diffusion model-based framework for deconstructing and modeling population-level pathology image data. Our solution enables multi-scale dynamic comparative visualization through an interactive interface, providing clinicians with an efficient tool for histopathological interpretation.

2.2 Large-Scale Image Analysis

Large-scale image visualization techniques play a critical role in analyzing massive image data. By integrating multi-level and multi-modal methods, these techniques significantly enhance the understanding of complex image content. Existing research mainly explores three directions: clustering analysis, deep learning models, and interactive visualization. In clustering-based hierarchical visualization, Xie et al.'s semantic co-embedding model projects images and keywords into 2D space but struggles with semantic extraction accuracy for complex medical images [25]. Bertucci et al.'s *DendroMap* employs interactive treemap layouts for hierarchical clustering, though its efficiency diminishes with ultra-large datasets [5]. Regarding deep learning applications, Niu et al.'s multi-modal multi-scale model provides rich feature representations but suffers from high training complexity [18]. For interactive techniques, Grotschla et al. proposed *AEye*, a visualization tool for image datasets that employs hierarchical and clustering methods to organize and display images, enabling users to intuitively explore and understand the structure and content of large-scale image datasets [9]. Rapp et al.'s moment-based representation enables efficient data compression and interaction but lacks precision in complex texture reconstruction [20]. Li et al. utilized pre-trained models for image captioning, though with limited applicability in specialized medical domains [14]. Additional contributions include a comprehensive review of large high-resolution display visualization [4], a graph-based method for image-text correlation analysis [10], and a systematic summary of image and video visualization methods [1]. For large-scale pathology image data, Tellez et al. proposed Neural Image Compression (NIC), a method that compresses pathology images into low-dimensional embeddings through unsupervised learning and then uses CNNs for analysis, achieving efficient visualization and interpretation [21].

While existing research has proposed various innovative solutions in large-scale image visualization, these approaches still face challenges in multi-scale navigation and real-time comparison of ultra-high-definition pathology images. Our proposed pyramid-based hierarchical loading and sliding comparison techniques aim to address these limitations by solving key problems in pathology image deconstruction, including multi-scale display and efficient interaction.

3 CLINICAL MOTIVATION

Histopathological examination is the clinical gold standard for diagnosing and evaluating a wide range of cancers, including colorectal, breast, gastric, liver, and lung cancers. Regardless of cancer type, pathological analysis typically involves tissue acquisition via resection or biopsy, followed by fixation, sectioning, and H&E staining to produce whole-slide pathology images. Despite biological variations among cancers, pathology image analysis tasks—such as tissue segmentation, component quantification, and morphological evaluation—are structurally and visually consistent across types. Leveraging this consistency, our

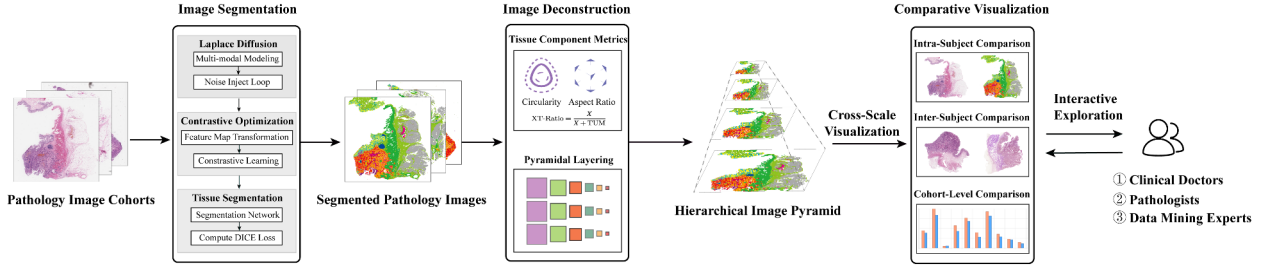


Fig. 2: The visual analytics pipeline of the pathology image visualization system composed of image segmentation, image deconstruction, and comparative visualization stages

goal is to develop a general-purpose visual analytics system for large-scale pathology image interpretation, applicable across different cancer domains.

4 DESIGN GOALS

This project stems from collaboration with experts in cancer pathology. Our core collaborators include two professionals: one is a clinical pathologist from a hospital, specializing in the pathological diagnosis of digestive system and soft tissue tumors, with over a decade of clinical experience (E1); the other is an academic researcher in pathology image analysis, whose expertise spans large-scale pathological models, AI-assisted diagnosis, and cancer prognosis analysis (E2). Through extensive discussions of clinical challenges, we defined the project's design objectives. Pathology images are characterized by ultra-high resolution and multi-scale complexity, requiring clinicians to repeatedly zoom in and out to examine both global structures and fine-grained details—a process that is both time-consuming and inefficient. To address this challenge, our work aims to develop an efficient and user-friendly visual analytics system to support clinical exploration and research on cancer pathology images. Accordingly, this paper presents a detailed examination of the following five key design dimensions:

DG1: Large-Scale Pathology Image Segmentation. Pathology images encapsulate rich morphological information about cellular and tissue structures distributed across hierarchical layers, involving diverse colors, densities, and tissue types. Clinicians require multi-level analysis of lesion areas, cellular compositions, and inter-tissue relationships to identify pathological features. Conventional whole-image observation inadequately resolves these complex structures, often limiting experts to superficial characteristics. Thus, ultra-large-scale image segmentation becomes imperative, enabling the decomposition of pathological content into distinct components to enhance analytical precision.

DG2: Quantitative Analysis of Pathological Tissues. Beyond visual information, pathology images contain quantifiable tissue characteristics such as lesion area, proportion, distribution density, and shape parameters. These metrics serve as critical evidence for disease progression assessment, classification, and clinical prognosis prediction. Traditional analysis relying on expert experience lacks systematic quantitative methodologies. Therefore, computational quantification of tissue proportions and morphological characteristics is essential to improve analytical objectivity.

DG3: Dynamic Comparison of Pathology Images. Pathology images exhibit gigapixel resolutions, where macroscopic views reveal tissue morphology, while microscopic levels display cellular structures and lesion characteristics. These diagnostic workflows necessitate frequent zooming to inspect global architectures and local pathological details. The enormous image size and complexity can result in time-consuming evaluations and potential diagnostic oversights. A flexible, dynamic, multi-scale comparison framework is thus required to facilitate comprehensive pathological analysis.

DG4: Multi-Level Visual Evaluation. Pathological studies extend beyond single-image analysis to comparative assessments across tissue components, individuals, and cohorts. For instance, single images may

exhibit unique organizational patterns under different pathological conditions, while inter-individual variations and cohort-level distribution patterns reveal systematic differences. Conventional methods confined to single-image observation lack effective multi-level analytical capabilities. Hence, cross-component, cross-individual, and cross-cohort visualization tools are essential for pathologists to gain deeper insights into lesions.

DG5: Efficient Interactive Exploration. The ultra-large-scale, ultra-high-resolution (gigapixel-level) characteristics of pathology images impose significant challenges on existing interaction paradigms. Constrained by data volume and resolution limitations, clinicians often resort to tedious manual comparisons, hindering efficient deep exploration. Consequently, there is an urgent need for seamless interaction mechanisms tailored to ultra-high-resolution pathology images, enabling rapid navigation, precise filtering, and in-depth analysis to substantially elevate diagnostic efficiency.

5 METHODS

5.1 Overall Workflow

As shown in Fig. 2, the overall workflow of our system includes pathology image segmentation, image deconstruction, and comparative visualization. In the first phase, we employ the Laplace diffusion model to segment the pathology images. Each image is decomposed into several pathological layers (i.e., tissue components), each labeled with a distinct color. In the second phase, we apply a multi-level pyramid structure to the segmented images to support smooth zooming interactions. Simultaneously, we compute the proportions and morphological indicators of different tissue components. In the third phase, users can interactively explore pathology images and quantitative indicators at multiple scales, thereby supporting researchers in conducting in-depth analysis and comparative evaluation of pathology images.

5.2 Pathology Image Segmentation

In this section, we introduce the Laplace Diffusion Model, called L-Diffusion, which is a new framework designed for efficient pathology image segmentation (DG1). The Laplace diffusion model enhances the distinguishability between different tissue component distributions, thereby improving segmentation accuracy and robustness in recognizing tail classes in pathology images. The process consists of three stages: Laplace diffusion, contrastive optimization, and tissue segmentation.

Firstly, Laplace diffusion is performed on the pathology image. For an original image x_0 , composed of N distinct components, we model it using N independent Laplace distributions, written as $q_1(x_0), q_1(x_0), \dots, q_N(x_0)$. During the diffusion process, the mean squared error (MSE) loss at the t -th step is calculated as:

$$\mathcal{L}_{MSE} = \mathbb{E}_{x_0, \epsilon, t} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right],$$

where ϵ denotes the added Laplace noise, and $\epsilon_\theta(x_t, t)$ is the predicted noise from the intermediate reconstructed feature map x_t using model parameters θ .

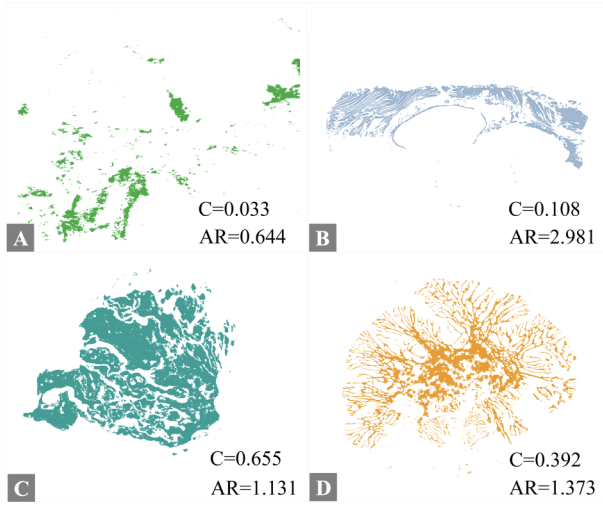


Fig. 3: Representative examples of tissue morphology quantification. Tissues with different morphological characteristics are labeled with distinct colors, and the corresponding C and AR values are displayed for each region.

Next, contrastive optimization is applied to the pixel latent vector. After T diffusion steps, each reconstructed feature map r_t is converted into a grayscale image $\hat{r}_t = \text{RGB2Gray}(r_t)$. Along the diffusion axis, each pixel corresponds to a vector $v \in \mathbb{R}^{1 \times T}$, referred to as the pixel latent vector. A contrastive learning loss function $\mathcal{L}_{CRT}^{k,k'}$ is applied:

$$\mathcal{L}_{CRT}^{k,k'} = -\log \frac{\text{Positive}(v_k^n, v_{k'}^n)}{\text{Positive}(v_k^n, v_{k'}^n) + \text{Negative}(v_k^n, v_{k'}^n)},$$

which encourages similarity between positive pairs $\text{Positive}(v_k^n, v_{k'}^n)$, and dissimilarity between negative pairs $\text{Negative}(v_k^n, v_{k'}^n)$.

Finally, we perform pathology tissue segmentation. By combining the MSE loss $\mathcal{L}_{MSE}$ and the contrastive loss $\mathcal{L}_{CRT}^{k,k'}$, the segmentation network $\mathcal{F}_{seg}(\cdot)$ generates the prediction y' as follows:

$$y' = \mathcal{F}_{seg}([f_1', f_2', \dots, f_t', \dots, f_T']).$$

The Dice loss between the predicted segmentation y' and the ground truth y is defined as:

$$\mathcal{L}_{Dice}(y', y) = 1 - \frac{2y' \cdot y}{y' + y}.$$

Through this two-stage training framework that integrates L-Diffusion and the segmentation network, pathology images are effectively decomposed into their constituent tissue components. We compared L-Diffusion with several mainstream models and achieved superior performance across multiple evaluation metrics, validating its high accuracy and robustness in pathological image segmentation.¹

5.3 Tissue Quantification

Following image segmentation, distinct pathological tissues are delineated into separate layers. Clinicians evaluate patient conditions by analyzing these tissue components, including their proportional distributions and morphological characteristics. Building upon this foundation, we developed a dual-path analytical approach: tissue proportion analysis via pixel ratio calculations and morphological characterization using shape factors (**DG2**).

¹Experimental details are provided in the supplementary material

Algorithm 1 Calculation of Aspect Ratio

Require:

$I_{m \times n}$: Binary Image Matrix ($I(i, j) \in \{0, 1\}$)

Ensure:

AR : Aspect ratio of the target region ($AR \geq 1$)

```

1: Initialization
2:  $[rows, cols] \leftarrow \text{size}(I)$ 
3:  $[X, Y] \leftarrow \text{meshgrid}(1 : cols, 1 : rows) \triangleright$  Generate Coordinate Grid
4: Calculate Centroid
5:  $A \leftarrow \sum_{i,j} I(i, j)$ 
6:  $c_x \leftarrow \sum_{i,j} j \cdot I(i, j) / A$ 
7:  $c_y \leftarrow \sum_{i,j} i \cdot I(i, j) / A$ 
8: Calculate Central Moments
9:  $X_c \leftarrow X - c_x$ 
10:  $Y_c \leftarrow Y - c_y$ 
11:  $\mu_{20} \leftarrow \sum_{i,j} X_c(i, j)^2 \cdot I(i, j)$ 
12:  $\mu_{02} \leftarrow \sum_{i,j} Y_c(i, j)^2 \cdot I(i, j)$ 
13:  $\mu_{11} \leftarrow \sum_{i,j} X_c(i, j) \cdot Y_c(i, j) \cdot I(i, j)$ 
14: Calculate Normalized Central Moments
15:  $\bar{\mu}_{20} \leftarrow \mu_{20} / A$ 
16:  $\bar{\mu}_{02} \leftarrow \mu_{02} / A$ 
17:  $\bar{\mu}_{11} \leftarrow \mu_{11} / A$ 
18: Calculate Eigenvalues of the Inertia Matrix
19:  $M \leftarrow \begin{bmatrix} \bar{\mu}_{20} & \bar{\mu}_{11} \\ \bar{\mu}_{11} & \bar{\mu}_{02} \end{bmatrix}$ 
20:  $\lambda_{1,2} \leftarrow \text{eig}(M) \triangleright \lambda_1 \geq \lambda_2$ 
21: Calculate Aspect Ratio
22: if  $\lambda_2 > 0$  then
23:    $AR \leftarrow \sqrt{\lambda_1 / \lambda_2}$ 
24: else
25:    $AR \leftarrow 1 \triangleright$  Degenerate to Circular
26: end if
27: return  $AR$ 

```

Tissue Proportion Quantification. To quantitatively evaluate the proportion of tissue components, we conducted interviews with domain experts and identified that in colorectal cancer pathology assessment, the necrosis tumor ratio (NTR) and tumor stroma ratio (TSR) serve as two critical prognostic indicators. Clinical evidence demonstrates that tracking dynamic changes in NTR and TSR before and after treatment can effectively evaluate tumor regression and therapeutic response. We further investigated the computational methodology of NTR and TSR, and found that both metrics can be expressed using the following unified formula:

$$\text{XTRatio} = \frac{X}{X + \text{TUM}}$$

Here, XTRatio represents the proportional contribution of a specific tissue component (NTR or TSR); X denotes the area of the target tissue component—Necrotic Epithelium (NEC) for NTR, or Stromal Tissue (STR) for TSR—and TUM refers to the tumor epithelium region, representing viable tumor cell populations.

This formulation provides a general computational framework for tissue component quantification. By substituting different values for X , we systematically quantify the distribution of various tissue components relative to the tumor epithelium, enabling comprehensive characterization of tumor microenvironment heterogeneity.

Tissue Morphology Quantification. Quantitative analysis of tissue morphology features originates from prior clinical knowledge in cancer pathology. For instance, benign lesions, metastatic lymph nodes, and normal glandular structures typically exhibit regular architectures, whereas invasive carcinomas often demonstrate highly irregular tissue patterns. Furthermore, existing studies corroborate that malignant tissues generally display higher boundary complexity compared to the smoother contours of benign lesions [19]. Research also indicates that in dermatologic imaging, basal cell carcinomas typically exhibit larger aspect ratios, whereas benign nevi tend toward circularity [3]. Malignant

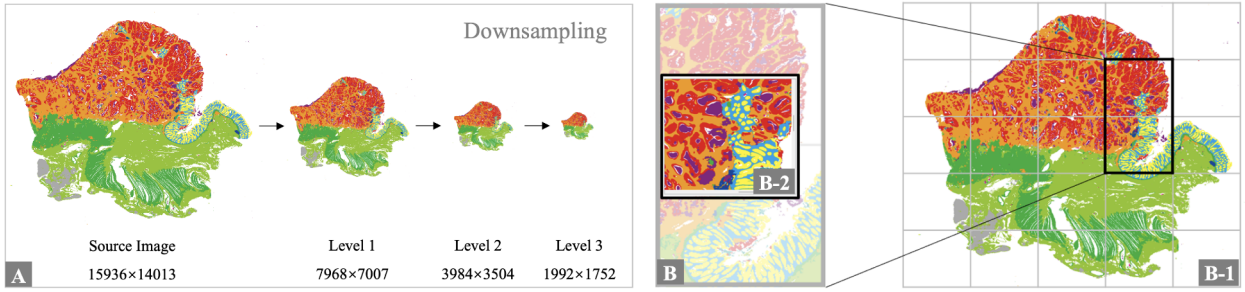


Fig. 4: Pyramid Layering involves (A) image pyramid compression followed by (B) cross-tile view rendering to enable multi-scale analysis.

nant lesions frequently manifest irregular, elongated morphologies or invasive growth patterns. Therefore, we define two quantitative metrics (circularity and aspect ratio) to systematically evaluate histopathological structural regularity, as follows:

Circularity. Circularity is a quantitative metric used to characterize how closely a 2D shape approximates a perfect circle, and also evaluates the regularity of its boundary. The formal definition is given by:

$$C = \frac{4\pi A}{P^2} \quad (1)$$

where A denotes the area of the shape, P represents its perimeter, and $C \in (0, 1]$ quantifies the circularity. Values approaching 0 indicate highly irregular or elongated morphologies, while $C = 1$ corresponds to a perfect circle.

Aspect Ratio. *Aspect Ratio (AR)* is a metric used to evaluate the morphology of lesioned tissue, calculated as the ratio of length to width of the tissue in pathology images, defined as:

$$AR = \frac{\text{Major Axis}}{\text{Minor Axis}} \quad (2)$$

Here, Major Axis is the longest axis of the minimum bounding ellipse fitted to the region, and Minor Axis is the shortest axis of this ellipse.

Fig. 3 illustrates four representative examples of quantitative morphological analysis in pathological tissues. Specifically: Fig. 3A shows scattered tissue morphology with relatively low values for both C and AR ; Fig. 3B displays elongated tissue exhibiting a small C value but a significantly large AR value; Fig. 3C demonstrates compact tissue with high values for both parameters; and Fig. 3D presents radial tissue morphology that also shows comparatively large C and AR values. Notably, while Fig. 3D appears visually more circular than Fig. 3C, its C value is actually lower. This discrepancy arises because C incorporates perimeter calculations, and the perimeter of radial structures increases substantially due to their branching patterns. As a result, circularity measurements can also serve as an effective metric for evaluating boundary continuity in pathological tissue analysis. The pseudocode for the aspect ratio calculation process is presented in Algorithm 2.

5.4 Pyramid Layering

In terms of pathology image visualization, since a single pathology image typically has dimensions ranging from tens of thousands to hundreds of thousands of pixels in width and height (with pixel counts in the billions), directly rendering full-resolution images poses prohibitively high computational and storage demands. To address this challenge, our system employs a pyramid-level storage architecture and tile-based loading strategy (DG3).

Image Pyramid Compression. The image pyramid architecture enables efficient visualization by pre-generating multi-resolution levels, where each successive level reduces resolution by 50% from the original base image. This hierarchical structure utilizes an image file directory index table to systematically organize all image levels, allowing rapid access to specific resolutions through offset addressing. During operation, the system dynamically determines the optimal resolution

level based on the viewport's zoom scale and efficiently loads corresponding image blocks. To maintain seamless visual transitions during zooming, the architecture implements a normalized coordinate system that recalculates and remaps the viewport center coordinates across different resolution levels, ensuring stable content anchoring. This approach significantly reduces computational complexity and optimizes memory usage by loading only the visible viewport area.

During thumbnail generation, the nearest-neighbor interpolation algorithm is employed, where each pixel in the target image directly inherits the original value from its closest corresponding pixel in the source image. This approach maintains processing efficiency while preventing interpolation artifacts, thereby ensuring strict color consistency within homogeneous tissue regions.

Tile-based Loading. The tile-based loading strategy employs dynamic rendering of only the currently visible image regions. When examining localized areas of pathology images, the system only loads and renders the image tiles within the current viewport, rather than the entire image. Fig. 4B presents the schematic diagram of the rendering process. This on-demand loading mechanism reduces memory usage depending on the current zoom level while improving interactive performance. The tile-based dynamic loading strategy is implemented through three key components:

Image Tiling. The original image is partitioned into fixed-size tiles (256×256 pixels) stored in a single file with offset-based indexing. Tile offsets across pyramid levels (e.g., full resolution and half resolution) are maintained independently to ensure efficient multi-level access.

Dynamic Loading. Only tiles covering the current viewport are loaded. When the viewport spans multiple tiles, the system calculates the required tile range, stitches them into a temporary canvas, and crops the target display area.

Cache Optimization. A multi-tier caching mechanism enhances performance. Loaded tiles are cached in memory for rapid access; cache hits bypass disk I/O, while misses result in tile loading and corresponding cache updates. The Least Recently Used (LRU) algorithm manages [6] cache eviction with a default limit of 200MB to prevent memory exhaustion. Additionally, prefetching anticipates adjacent tile demands based on user navigation patterns (e.g., panning or zooming), effectively reducing interaction latency and ensuring a smooth user experience.

5.5 Comparative Visualization

The design of the system's comparison functionality was finalized through multiple rounds of in-depth interviews and discussions with two senior pathology experts. It specifically addresses the three most common and critical comparison tasks encountered in routine clinical pathology practice: intra-subject, inter-subject, and cohort-level comparison. It should be noted that our goal is not to create complex or novel visualization techniques for deep pattern discovery, but to implement clear, efficient, and clinically practical comparison strategies that effectively alleviate the operational burden caused by frequent zooming and navigation during pathology image analysis (DG4, DG5).

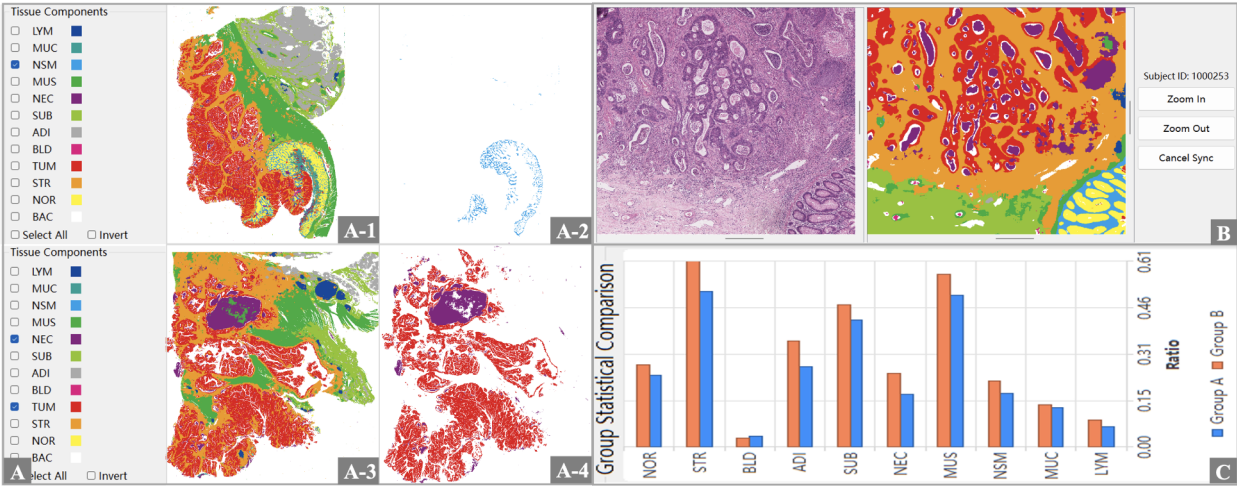


Fig. 5: Comparative Visualization includes: (A) Intra-Subject Comparison, (B) Inter-Subject Comparison, (C) Cohort-level Comparison.

Intra-Subject Comparison. Since clinical pathologists need to examine the spatial relationships among multiple tissue components within a single pathology image slice, we adopt a superposition approach that enables selective tissue highlighting. This design helps pathologists focus on specific regions of interest while reducing visual clutter. The system features an interactive channel selection interface, enabling dynamic visualization control for diverse tissue types. Leveraging diffusion model-based deconstruction technology, pathology images are decomposed into multi-layered tissue planes. A high-contrast color scheme assigns distinctive identifiers to each tissue type, with checkbox controls allowing real-time display customization.

Fig. 5A demonstrates the comparative visualization between the original segmented image (Fig. 5A-1) and the NSM-highlighted view (Fig. 5A-2). When multiple tissue components overlap in the segmented pathology image, as seen in Fig. 5A-1, it becomes challenging to observe the target tissue clearly. Selecting NSM in the control panel triggers automatic highlighting (Fig. 5A-2), effectively removing visual clutter caused by surrounding tissue types. The system also calculates real-time area ratios for each tissue and generates comparative histograms. By default, only the top five most prevalent tissues are displayed (Fig. 5A-3). Upon selecting specific tissues via the color channel interface, the histogram dynamically updates to show sorted ratios of selected tissues, excluding the white background. This design addresses interface congestion when more than 12 tissue types are present and significantly enhances exploratory efficiency. Users can further customize the view by adding or removing tissue categories of interest. Fig. 5A-3 illustrates multi-tissue co-visualization. By simultaneously selecting NEC (Necrosis) and TUM (Tumor Epithelium), the interface highlights both tissues within the segmented image (Fig. 5A-4), supporting detailed analysis of necrotic distribution within tumor regions.

The multi-channel selection approach enables dynamic composition of multiple tissue types, allowing users to emphasize structures of interest while suppressing visual noise from irrelevant components. This method effectively mitigates information occlusion issues in original deconstruction maps, helping clinicians better identify latent pathological patterns.

Inter-Subject Comparison. Since clinical pathologists need to compare the structural and morphological differences between the original stained images and the deconstructed segmented images, we adopt a juxtaposition design to enable intuitive side-by-side visualization. The system also supports synchronized zooming to enhance comparison accuracy and efficiency, as seen in Fig. 5B. The original image presents the natural morphology of the tissue, while the segmented image clearly delineates the distribution and boundaries of different

tissues through color labeling. Users can click the "Detailed Analysis" button on any image group to access an in-depth analysis interface. Within this interface, the subject ID is displayed by default, and both images support independent zooming and panning. When the "Sync Zoom" function is enabled, the two images synchronize in real time, ensuring precise alignment of corresponding regions for convenient localized comparison.

Cohort-level Comparison. Given that clinical pathologists also need to analyze tissue composition differences across various patient subtypes, we implemented a cohort-level comparison approach that combines sliding navigation with tissue composition statistics, enabling efficient cohort-level browsing and identification of distribution patterns. The system enables efficient comparative analysis by simultaneously displaying pathology images in dual windows. Users can horizontally swipe to quickly switch between samples within a group for convenient data review. In practical applications, multiple pathology images from the same patient can be grouped to analyze dynamic changes in tissue distribution and composition within an individual. Alternatively, patients can be grouped by disease type to compare histological differences across various pathological patterns, facilitating disease research. To enhance inter-group comparison, the system automatically calculates and displays statistical charts of average XT-Ratio metrics (below the tissue selection panel), as seen in Fig. 5C, which dynamically correspond to selected tissue types for interactive analysis.

6 EVALUATION

To assess the effectiveness and practical applicability of our visual analytics system, we conducted a case study using real-world clinical pathology data. The evaluation focused on verifying the system's performance in segmenting tissue components, quantifying pathological features, and supporting multi-level visual analysis.

For this purpose, we used a dataset comprising 186 pathologically confirmed colorectal cancer cases (TNM stage I–III) provided by the First Affiliated Hospital of Zhejiang University. Colorectal cancer was selected as a representative case due to its well-characterized histopathological patterns and the availability of high-quality annotated data. Two domain experts (E1 and E2) were invited to participate in the system evaluation. We first verified its ability to reflect established pathological knowledge, followed by expert-led assessments of usability and effectiveness in real-world clinical tasks.

Notably, the system architecture was designed with generalizability in mind. Its core modules—including tissue segmentation, component quantification, and multi-level comparison—were developed without reliance on cancer-type-specific assumptions, enabling application to pathology images across a range of cancer domains.

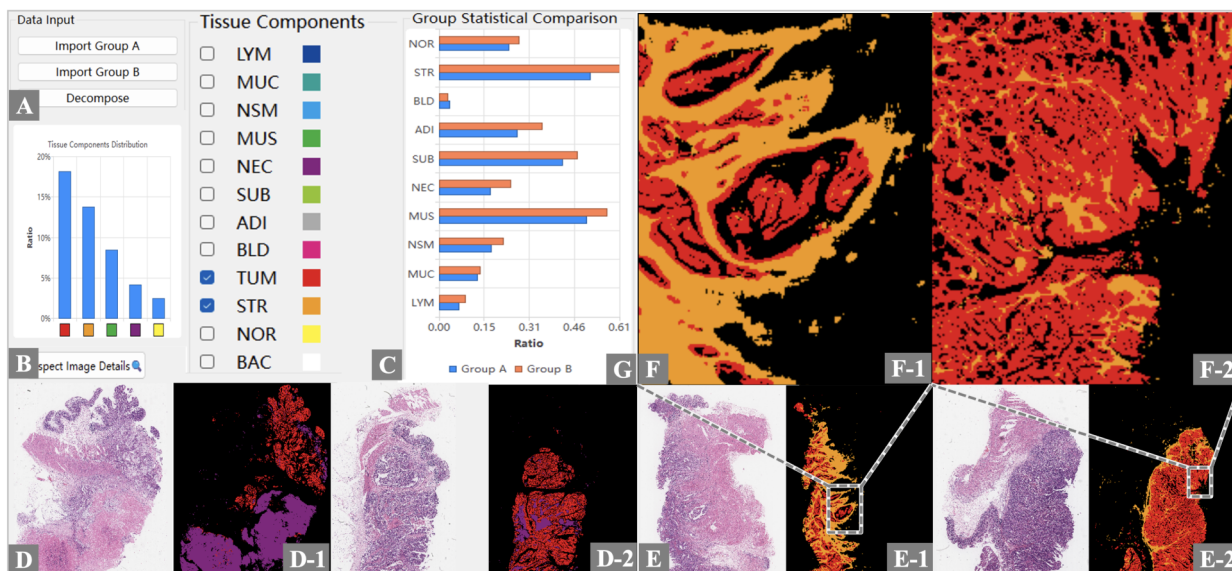


Fig. 6: Pathology knowledge validation process: (A) data input, (B) tissue proportion histogram, (C) tissue components selection, (D) NEC vs. TUM comparison, (E) STR vs. TUM comparison, (F) magnified view, and (G) group statistical comparison.

6.1 Case1: Pathology Knowledge Validation

Our clinical data were divided into two groups: Group A (good prognosis, no recurrence or metastasis after treatment) and Group B (poor prognosis, with recurrence or metastasis). As shown in Fig. 6, we first loaded the pathology images via the data import module (Fig. 6A) and performed model segmentation by clicking the “Decompose” button. Subsequently, we used a sliding window to identify representative images with high proportions of TUM, STR, and NEC in both groups, as visualized in the tissue proportion histogram (Fig. 6B).

Comparative histomorphological analysis revealed critical differences in necrosis distribution patterns [26]. In the NEC vs. TUM comparison (Fig. 6D), Group A exhibited sparse and low-proportion necrosis (Fig. 6D-1), whereas Group B showed clustered and extensive necrosis (Fig. 6D-2), suggesting a correlation between necrosis and tumor aggressiveness. A closer examination revealed that the distribution pattern of necrotic tissue in Group B was not only more extensive but also more structurally heterogeneous. E1 noted that the irregular border configurations may be linked to tumor hypoxia and treatment resistance.

Stromal distribution patterns emerged as another key prognostic differentiator. The STR vs. TUM comparison (Fig. 6E-F) demonstrated significantly greater stromal coverage in Group B, evidenced by extensive orange regions in Fig. 6E-1. Detailed inspection via the “Inspect Image Details” function (Fig. 6F-1 and F-2) revealed that STR in Group B formed a “physical barrier” enveloping TUM, while Group A displayed minimal STR without encapsulation. E1 pointed out that this encapsulating pattern may serve as a protective mechanism for tumor cells, preventing immune cell infiltration or therapeutic agent penetration, which could contribute to treatment resistance and poor prognosis. This supports the hypothesis that stroma may impede drug delivery to tumor cells, consistent with prior findings [26].

Furthermore, the “Group Statistical Comparison” bar chart (Fig. 6G) confirmed that Group A had significantly lower NTR and TSR than Group B, aligning with existing studies [26]. TSR, as a quantitative metric, holds high clinical relevance for cancer analysis. E1 noted that TSR might serve as a potential prognostic indicator, as patients with high TSR values tended to exhibit more aggressive tumor characteristics. E1 further stated: “This comparative approach enables intuitive and efficient multi-indicator analysis, facilitating the inclusion of additional metrics in future studies. Future work can focus on integrating

additional features, such as immune cell infiltration patterns and extracellular matrix composition.” Building on these observations, E1 maintained that the combined examination of necrosis patterns, stromal encapsulation, and quantitative ratios (e.g., TSR, NTR) provides a more nuanced understanding of tumor progression than any single metric alone. He further noted that such multidimensional exploration, enabled by simultaneous visualization and statistical summaries, facilitates the formulation of mechanistic hypotheses.

6.2 Case2: Pathology Image Analysis

In this case study, experts E1 and E2 independently operated our system to analyze pathology images. E1 began by quickly scanning the whole-slide image using a sliding window approach to identify regions of interest. By examining the original image alongside the segmented image and tissue proportion histogram, shown in Fig. 7A–C, E1 confirmed the effectiveness of the system’s high-contrast color scheme. The clear visual differentiation between tissue types provided an intuitive interface that facilitated rapid identification of pathological features. Noting its adherence to pathological conventions, E1 highlighted that normal muscle and mucosal tissues were displayed in green, while tumor tissues appeared in red, ensuring easy recognition.

E1 specifically identified the dark blue areas as lymphoid tissue and selected these regions along with surrounding tissues for detailed examination (Fig. 7B-1). This step allowed E1 to focus on potential diagnostic clues and evaluate tissue organization more effectively. Analysis of the tissue morphology quantification bar charts (Fig. 7C-1, C-2) revealed that while the lymphoid tissue exhibited significantly higher circularity values compared to adjacent tissues, their AR values remained similar. E2 noted that this finding demonstrates the system’s capability to incorporate multiple morphological parameters for tissue characterization. E1 then zoomed in for closer inspection, leveraging the system’s synchronized zoom functionality between the original and segmented images. He seamlessly navigated between magnification levels to conduct a detailed analysis of cellular morphology and tissue architecture. However, E1 noted the system’s current limitation in providing finer-grained tissue analysis at the cellular or subtype level, such as distinguishing between T-cells and B-cells within lymphoid tissue, which could provide valuable diagnostic information. To address this, E1 suggested further enhancing the system’s diagnostic potential by enabling automatic differentiation of immune cell subtypes.

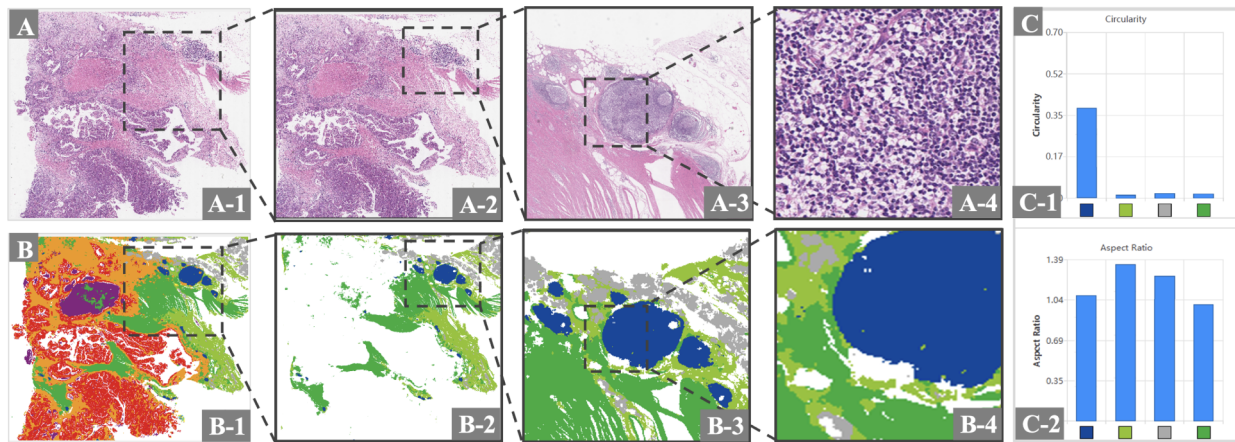


Fig. 7: Pathology image exploration workflow: Enlarged visualization of target regions (A-1 to A-4, B-1 to B-4) followed by quantitative analysis of tissue metrics (C-1, C-2).

E2 adopted a more data-driven analytical approach, focusing on the system’s analytical capabilities and interactive features. Unlike E1, who primarily concentrated on pathological content discovery, he leveraged the system’s quantitative tools for in-depth analysis. He particularly valued the system’s robust tissue quantification and distribution visualization features, as well as its ability to smoothly display images at various magnification levels, similar to Fig. 7. These features allowed E2 to investigate subtle differences in tissue composition that might not be immediately apparent in raw images. E2 suggested additional functionalities to further enhance pathological research, including the ability to calculate custom metrics using user-defined formulas and to perform localized statistical analyses tailored to specific study needs. For example, he proposed quantifying tissue proportions around tumor boundaries to better assess invasion patterns, which are critical for determining tumor progression and patient prognosis.

Additionally, E2 emphasized that implementing more sophisticated metrics to precisely quantify the tumor microenvironment would significantly benefit pathological studies. By enabling deeper insights into tumor–stroma interactions, these advancements could refine diagnostic accuracy and improve personalized treatment strategies. Furthermore, during interaction, E2 expressed a desire to highlight specific pathological details, such as nuclear orientation in pathology images (Fig. 7A-4). He therefore suggested adding a real-time annotation feature, enabling researchers to directly mark regions of interest within the system for easier analysis and discussion. These enhancements would collectively strengthen the system’s role as a powerful tool for advancing digital pathology.

6.3 Expert Feedback

Experts unanimously recognized the overall functionality and clinical value of the system, affirming that its multimodal tissue quantification method and dynamic visual interactive design effectively overcome the challenges of analyzing ultra-large-scale pathology images, making it an intelligent platform bridging clinical diagnosis and cutting-edge research.

E1 highly praised the system’s practicality, particularly emphasizing that multimodal tissue quantification is a key innovation. In particular, the intuitive visualization of the spatial relationship between STR and TUM is highly beneficial for tumor prognosis assessment. However, he pointed out that the system currently lacks cell subtype analysis (e.g., distinguishing between T/B lymphocytes), a critical function for tumor microenvironment research. He suggested incorporating cell subtype recognition in future iterations to enhance clinical applications. Additionally, E1 noted that clinicians have limited time for research and typically use the system at night or from home. Therefore, he

recommended developing a web-based remote access version to better support multicenter research collaboration.

E2 highlighted the system’s outstanding performance in group comparative analysis, making it a reliable quantitative tool for tumor prognosis research. He specifically emphasized that the dynamic visual interactive design significantly enhances the exploratory capability of research data. Moreover, E2 suggested adding user-defined analysis formula functionality to improve the system’s flexibility and expandability. He further proposed that future enhancements could support attribute-based localized region analysis, such as evaluating tissue distribution within 5 micron and 10 micron tumor margins to assess tumor spread. Additionally, analyzing the lymphocyte infiltration gradient within specific tumor boundary regions could further improve understanding of the tumor microenvironment, providing stronger support for precision medicine.

Operation	Time (s)
Tissue proportions and XT-Ratio	0.547
Morphology quantification	0.680
Overlay of two tissue layers	0.475
Six-level image pyramid	8.677

Table 1: Computation time of pathology image deconstruction.

7 PERFORMANCE

The visualization system is specifically designed for standard clinical environments, compatible with mainstream consumer laptops (Windows 11 OS / 13th Gen Intel Core i5-13500H processor / 32 GB RAM). The diffusion model was pre-trained on a workstation with 8×NVIDIA A800 GPUs, utilizing approximately 490,000 training samples sourced from multiple institutions and publicly available resources, including PubMed databases, pathology textbooks, and atlases, etc. For practical deployment, the system only requires an 8 GB VRAM device to perform inference. Benchmark tests show that fine-grained segmentation of a single 800 MB pathology image takes approximately 5 minutes.

The system achieves sub-second processing time for statistical operations and second-level latency for pyramid generation per image, as presented in Table 1. All intermediate statistical results and pyramid files are instantly written to disk, eliminating redundant computations for subsequent data loading and significantly enhancing processing efficiency.

8 DISCUSSION AND LIMITATION

Large and multi-scale pathology image analysis plays a crucial role in cancer research. In this work, we focus on addressing the practical needs of clinical experts by integrating diffusion models with visual analytics, enabling precise quantification of diverse tissue components within pathology images. The system incorporates standard visualization techniques (e.g., bar charts, zooming and panning interactions) to ensure usability and intuitive interpretation in real-world clinical settings. The algorithmic integration within the system follows a modular design, allowing domain experts to replace or embed different segmentation models as needed. This design ensures flexibility and facilitates adaptation to varying analytical tasks. In addition, the current implementation supports pathology image analysis for a single cancer type and can be further customized to accommodate other cancer types (e.g., lung cancer, breast cancer) based on expert demands.

While the system effectively supports expert-driven exploration of pathology images, several limitations remain. First, the current version is implemented as a standalone desktop application, which limits accessibility and real-time collaborative use. This could be addressed by developing a web-based version supported by a high-performance backend for multi-user deployment. Second, due to the sensitivity of pathology image data, the system prioritizes offline expert analysis. To facilitate secure multi-institutional collaboration, future development should incorporate enhanced data protection mechanisms, such as encrypted computing or federated learning frameworks.

Moreover, while the current system emphasizes simplicity and operational efficiency to meet clinical workflow requirements, we acknowledge that more advanced visual paradigms—such as tissue layer superposition and multi-modal overlays—may be valuable in research-oriented scenarios. Such capabilities are not included by default, but can be flexibly incorporated based on specific expert needs and study contexts to support more sophisticated comparative analysis.

9 CONCLUSION

This paper presents an intelligent visual analytics system for exploring large and multi-scale pathology images, combining diffusion models and interactive visualization to enhance clinical pathology analysis. The system leverages Laplace Diffusion for accurate tissue segmentation and employs pyramid layering for efficient rendering of gigapixel-resolution images. By enabling multi-scale dynamic comparison (intra-subject, inter-subject, and cohort-level) and quantitative morphological analysis, the system improves diagnostic objectivity and efficiency. Evaluations on real-world colorectal cancer datasets and feedback from clinical pathologists confirm its alignment with medical knowledge and practical utility.

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